Exp. No: 2

Electric Field and Electric Flux Density in a Sphere Full of Charges with Varying Radius

Aim:

To write and validate MATLAB programs for electric field intensity (E) and electric flux density (D) for specified spherical charge distributions.

Objectives:

1. To derive expressions for electric field intensity (E) and electric flux density (D) inside and outside a uniformly charged sphere using Gauss's law.

2. To plot the variation of E and D with radial distance for varying sphere radius and charge density.

3. To validate the simulated results against theoretical calculations for different radii.

Program:

Test case (i) Determination and plotting of electric field and electric flux density in sphere full of charges with varying radius

```
clc;

clear;

q=input('Enter the value of q=');

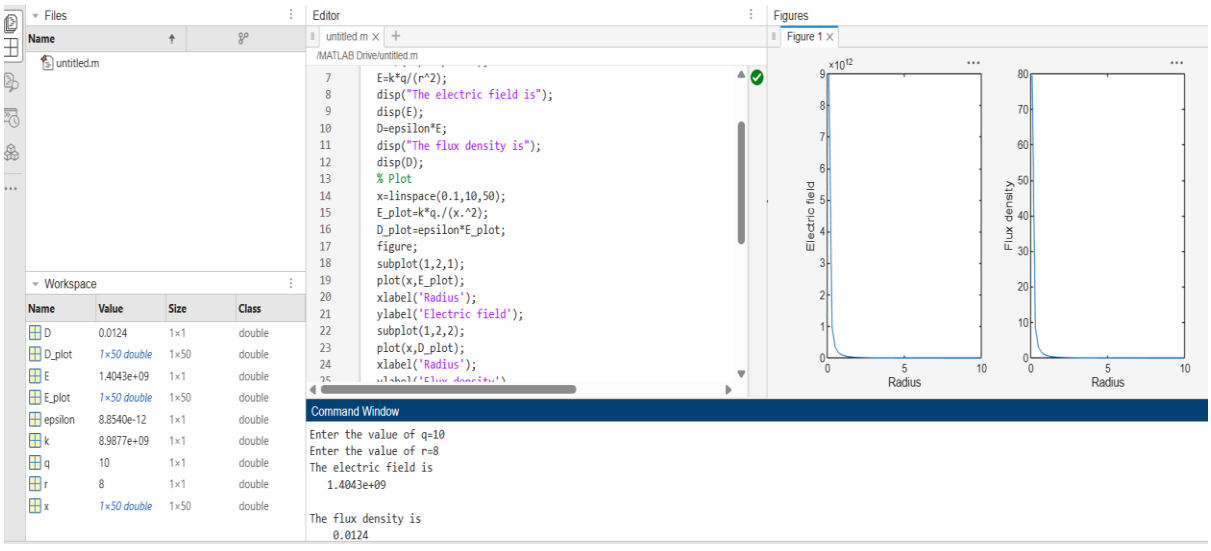
r=input('Enter the value of r=');

epsilon=8.854e-12;

k=1/(4*pi*epsilon);

E=k*q/(r^2);
```

```
disp("The electric field is");
disp(E);
D=epsilon*E;
disp("The flux density is");
disp(D);
% Plot
x=linspace(0.1,10,50);
E_plot=k*q./(x.^2);
D_plot=epsilon*E_plot;
figure;
subplot(1,2,1);
plot(x,E_plot);
xlabel('Radius');
ylabel('Electric field');
subplot(1,2,2);
plot(x,D_plot);
xlabel('Radius');
ylabel('Flux density')
```



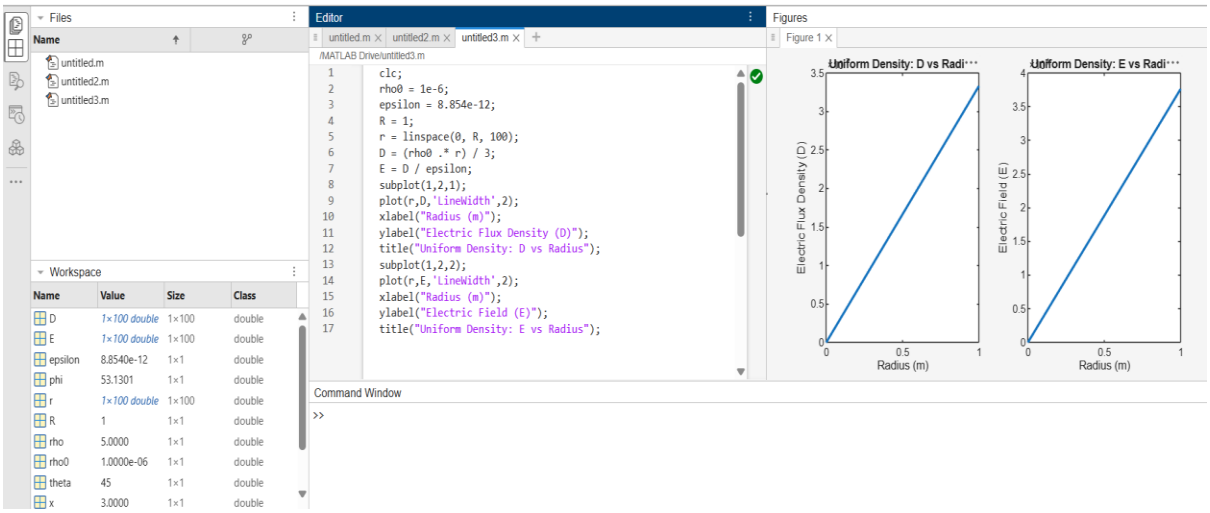
Test case (ii)

Derive and plot the electric field (E) and electric flux density (D) inside a uniformly

charged sphere with charge density ρ_0

Program:

```
clc;
rho0 = 1e-6;
epsilon = 8.854e-12;
R = 1;
r = linspace(0, R, 100);
D = (rho0 .* r) / 3;
E = D / epsilon;
subplot(1,2,1);
plot(r,D,'LineWidth',2);
xlabel("Radius (m)");
ylabel("Electric Flux Density (D)");
title("Uniform Density: D vs Radius");
subplot(1,2,2);
plot(r,E,'LineWidth',2);
xlabel("Radius (m)");
ylabel("Electric Field (E)");
title("Uniform Density: E vs Radius");
```



Test case (iii) For a sphere with a linearly increasing charge density $\rho_v = \rho_0 r$, derive and plot the

electric field (E) and electric flux density (D)

Program:

```
clc;

rho0 = 1e-6;

epsilon = 8.854e-12;

R = 1;

r = linspace(0, R, 100);

D = (rho0 .* r.^2) / 4;

E = D / epsilon;

subplot(1,2,1);

plot(r,D,'LineWidth',2);

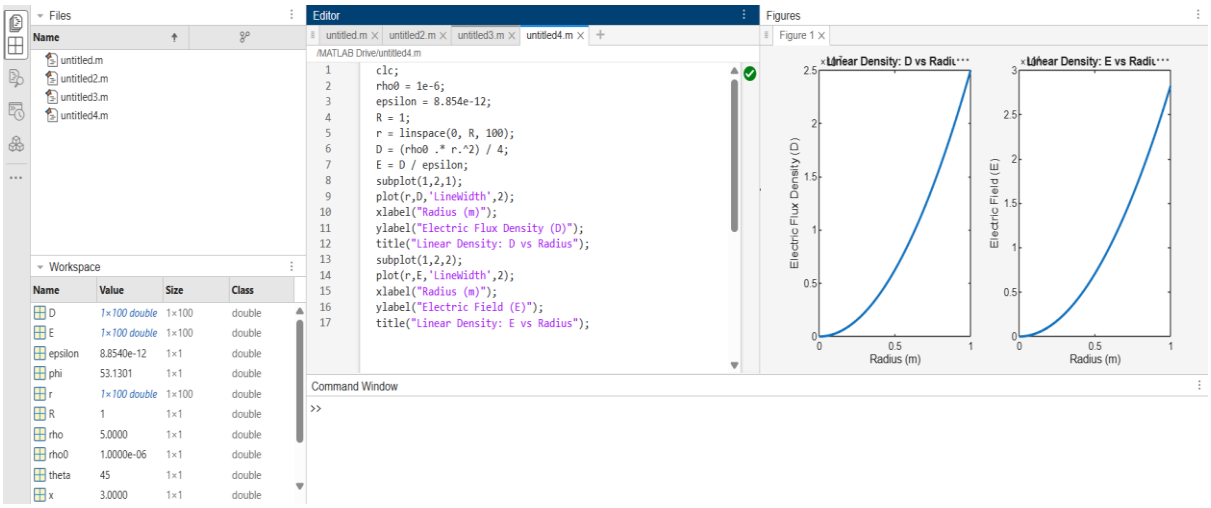
xlabel("Radius (m)");

ylabel("Electric Flux Density (D)");

title("Linear Density: D vs Radius");

subplot(1,2,2);
```

```
plot(r,E,'LineWidth',2);
xlabel("Radius (m)");
ylabel("Electric Field (E)");
title("Linear Density: E vs Radius");
```



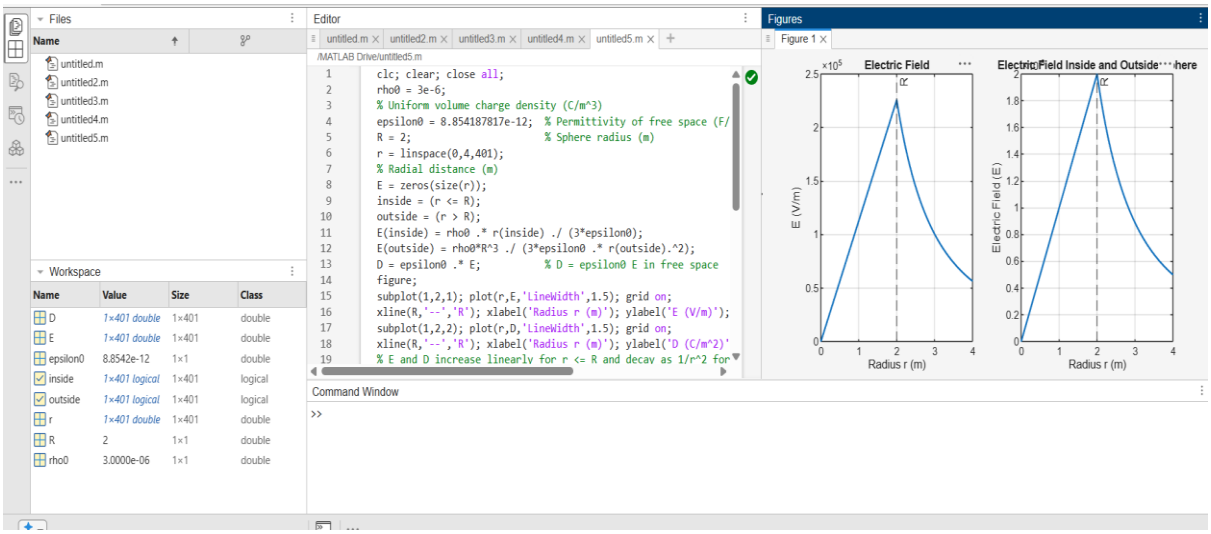
Test case (iv):

For a uniformly charged solid sphere of radius R = 2 m and volume charge density $\rho_v = 3\times10^{-6}$ C/m^3, calculate and plot E and D inside and outside the sphere.

Program:

```
clc; clear; close all;
rho0 = 3e-6;
% Uniform volume charge density (C/m^3)
epsilon0 = 8.854187817e-12;  % Permittivity of free space (F/m)
R = 2;                  % Sphere radius (m)
r = linspace(0,4,401);
 % Radial distance (m)
E = zeros(size(r));
inside = (r <= R);
```

```
outside = (r > R);
E(inside) = rho0 .* r(inside) ./ (3*epsilon0);
E(outside) = rho0*R^3 ./ (3*epsilon0 .* r(outside).^2);
D = epsilon0 .* E;          % D = epsilon0 E in free space
figure;
subplot(1,2,1); plot(r,E,'LineWidth',1.5); grid on;
xline(R,'--','R'); xlabel('Radius r (m)'); ylabel('E (V/m)'); title('Electric Field');
subplot(1,2,2); plot(r,D,'LineWidth',1.5); grid on;
xline(R,'--','R'); xlabel('Radius r (m)'); ylabel('D (C/m^2)'); title('Electric Flux Density');
% E and D increase linearly for r <= R and decay as 1/r^2 for r > R.
ylabel("Electric Field (E)");
title("Electric Field Inside and Outside Sphere");
```



Rubrics:

S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks Given	Marks Obtained
1	Select the appropriate charge distribution model.	Select the appropriate charge distribution model.	Correctly distinguishes point, uniform volume and specified non-uniform charge models and their assumptions.	Selects the correct model with minor clarification needed.	Shows partial understanding of the charge model.	Uses an unsuitable model or assumptions.	25	2.5	
2	Implement E and D calculations in MATLAB.	Implement E and D calculations in MATLAB.	Correctly calculates electric field E and flux density D with consistent SI units.	Calculations are mostly correct with minor coding or unit errors.	Some calculations work but require assistance.	Unable to obtain valid E and D values.	25	2.5	
3	Analyze radial variation and validate results.	Analyze radial variation and validate results.	Plots the correct inside/outside or stated distribution behaviour and verifies against theory.	Plots and validation are mostly correct with minor discrepancies.	Plots or theoretical validation are incomplete.	Plots are incorrect or results are not validated.	30	3	
4	Interpret and document findings.	Interpret and document findings.	Explains the effect of radius and charge distribution and presents labelled plots/tables.	Interpretation is generally correct with minor omissions.	Interpretation is limited or documentation incomplete.	Cannot interpret or document results.	20	2	
Total							**100**	**10**	